Aarhus University

Computer Science

Numerical Linear Algebra

Handin 8

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Consider the matrix

$$A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

(a) Characteristic Polynomial

Determine the characteristic polynomial $\det(A - \lambda I_3)$ of A.

$$A - \lambda I_3 = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} = \begin{bmatrix} 3-\lambda & 0 & 0 \\ 0 & 1-\lambda & 1 \\ 0 & 1 & 1-\lambda \end{bmatrix}$$

$$\det(A - \lambda I_3) = (3-\lambda) \cdot \begin{bmatrix} 1-\lambda & 1 \\ 1 & 1-\lambda \end{bmatrix} - 0 \cdot \begin{bmatrix} 0 & 0 \\ 0 & 1-\lambda \end{bmatrix} + 0 \cdot \begin{bmatrix} 0 & 1-\lambda \\ 0 & 1 \end{bmatrix} = (3-\lambda) \cdot ((1-\lambda) \cdot (1-\lambda) - (1 \cdot 1))$$

Definition 21.12 [1]

$$(3-\lambda) \cdot ((1-\lambda) \cdot (1-\lambda) - (1 \cdot 1)) = (3-\lambda) \cdot (\lambda^2 - 2\lambda)$$

Conclusion:

$$\det(A - \lambda I_3) = (3-\lambda) \cdot (\lambda^2 - 2\lambda)$$

(b-c) Some Eigenvalues

Show that 0 and 3 are eigenvalues of A and other values are welcome too.

The eigenvalues λ can be found by finding the solutions to the characteristic polynomial from (a), since any λ that results in the polynomials being 0 is an eigenvalue for A.
This means solving:

$$(3-\lambda) \cdot (\lambda^2 - 2\lambda) = 0$$

(With a hint from MatLab) this can formulated to finding a solution to:

$$3 - \lambda = 0 \quad \text{or} \quad \lambda^2 - 2\lambda = 0$$

trivially a solution is $\lambda = 3$, making 3 an eigenvalue for A:

$$3 - \lambda = 0 = 3 - 3$$

For the other one $\lambda^2 - 2\lambda$, the quadratic formula can be used
to find the roots for $1 \cdot \lambda^2 - 2 \cdot \lambda + 0 = 0$.

$$\lambda = \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \cdot 1 \cdot 0}}{2 \cdot 1} = \frac{2 \pm \sqrt{4}}{2} = \frac{2 \pm 2}{2}$$

This leaves two solutions

$$\frac{2+2}{2} = 2 \quad \text{and} \quad \frac{2-2}{2} = 0$$

which means the the eigenvalues for A is $[3, 2, 0]$

(d) Basis using Eigenvectors

Determine a basis of $\mathbb{R}^3$ consisting of eigenvectors of A.

Proposition 21.5 [1], says that if the eigenvalues of A are unique and there is a set of eigenvectores for A, then the eigenvectores constitues a basis for A.
Eigenvectores can be found by solving for every v in $(A - \lambda I_3)v = 0$ using the eigenvalues $\lambda \in [3, 2, 0]$

(e) Panic?!

Write $A = V\Lambda V^{-1}$ for suitable V and diagonal Λ.

(f) Witness me! - Said Donald Duck Calmly

Use item (e) to determine A^k for all k.

Appendix

References

[1] Andrew Swan. *Numerisk Linær Algebra*. Institut for Matematik, Aarhus Universitet, 2025.